

Q1: 16 March (Shift 2) - Single Correct

Let $A = \{2, 3, 4, 5, \dots, 30\}$ and ' $\simeq$ ' be an equivalence relation on $A \times A$, defined by $(a, b) \simeq (c, d)$, if and only if $ad = bc$. Then the number of ordered pairs which satisfy this equivalence relation with ordered pair $(4, 3)$ is

equal to :

(1) 5

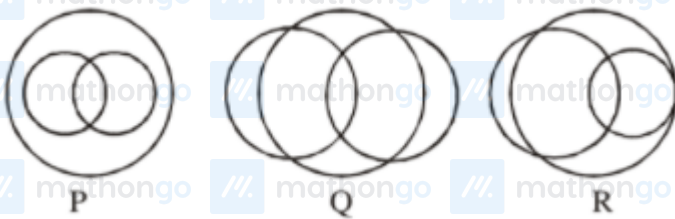
(2) 6

(3) 8

(4) 7

Q2: 17 March (Shift 1) - Single Correct

In a school, there are three types of games to be played. Some of the students play two types of games, but none play all the three games. Which Venn diagrams can justify the above statement?



(1) P and Q

(2) P and R

(3) None of these

(4) Q and R

Q3: 18 March (Shift 2) - Single Correct

Define a relation R over a class of $n \times n$ real matrices A and B as " ARB iff there exists a non-singular matrix P such that $PAP^{-1} = B$ ". Then which of the following is true?

(1) R is symmetric, transitive but not reflexive,

(2) R is reflexive, symmetric but not transitive

(4) R is reflexive, transitive but not symmetric

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

Answer Key**Q1 (4)****Q2 (3)****Q3 (3)**

Q1 (4)

$$A = \{2, 3, 4, 5, \dots, 30\}$$

$$(a, b) \simeq (c, d) \Rightarrow ad = bc$$

$$(4, 3) \simeq (c, d) \Rightarrow 4d = 3c$$

$$\Rightarrow \frac{4}{3} = \frac{c}{d}$$

$$\frac{c}{d} = \frac{4}{3} \quad \&c, d \in \{2, 3, \dots, 30\}$$

$$(c, d) = \{(4, 3), (8, 6), (12, 9), (16, 12), (20,$$

$$15), (24, 18), (28, 21)\}$$

$$\text{No. of ordered pair} = 7$$

Q2 (3)

$A \cap B \cap C$ is visible in all three venn diagram

Hence, Option (3)

Q3 (3)

A and B are matrices of $n \times n$ order & ARB iff

there exists a non singular matrix P ($\det(P) \neq 0$)

such that $PAP^{-1} = B$

For reflexive $ARA \Rightarrow PAP^{-1} = A \dots (1)$ must be true

for $P = I$, Eq.(1) is true so 'R' is reflexive For symmetric $ARB \Leftrightarrow PAP^{-1} = B \dots (1)$ is true

for BRA iff $PBP^{-1} = A \dots (2)$ must be true $\therefore PAP^{-1} = B$

$$P^{-1}PAP^{-1} = P^{-1}B$$

$$IAP^{-1}P = P^{-1}BP$$

$$A = P^{-1}BP$$

$$\text{from (2) \& (3) } PBP^{-1} = P^{-1}BP$$

can be true some $P = P^{-1} \Rightarrow P^2 = I$ ($\det(P) \neq 0$)

So 'R' is symmetric For transitive

$$ARB \Leftrightarrow PAP^{-1} = B \dots \text{is true}$$

Hints and Solutions

MathonGo

$BRC \Leftrightarrow PBP^{-1} = C \dots$ is true

now $PPAP^{-1}P^{-1} = C$

$P^2 A (P^2)^{-1} = C \Rightarrow ARC$

So 'R' is transitive relation

$\Rightarrow$ Hence R is equivalence